

NeoGravity Verification Suite

Companion computational materials for **The Theory of NeoGravity & Ether Dynamics**

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Version 4.2 · 17 September 2026. Supersedes version 4.1 (14 September 2026), which superseded 4.0 (6 September 2026), 3.1 and 2.0. **Twenty-seven claims**; version 4.0 carried nineteen. The suite prints its own version history when it runs, so a reviewer who ran an earlier release should start the program and read the header.

Where every claim comes from

Each of the twenty-seven prints the Part of the series it belongs to, the paper that deposits it, the numbered section and, where there is one, the subsection, together with what it recomputes, which of the five instruments it belongs to, and its epistemic standing. A reader who wants to check a claim against the paper behind it can go straight there.

```
=====
CLAIM 1 -- Hydrostatic equilibrium of the Ether Plenum
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  Part III. The Law
  Gravity as Hydrodynamics, Paper 11
  Section 3. Deriving the Index | 3.2 Hydrostatic Equilibrium
  Recomputes: rho(r) = rho_0 exp(3GM/rc^2) from the inward load under
                hydrostatic equilibrium
  The Plenum, and its constants | DERIVED
=====
```

The register those lines come from was **generated from the corpus rather than typed**: paper titles from the series concordance, section titles read out of the papers themselves, the Parts read out of the Series in Eleven Parts block. It is frozen into the one deposited file, and a companion instrument, `check_sources.py`, reads the papers back and verifies all twenty-seven rows character for character, so a renamed section is caught rather than left citing a place that is no longer there. The full register is reproduced in `claim_register.txt`.

The five instruments

They partition the twenty-seven exactly, and the program refuses to finish if they stop doing so.

Instrument	Claims
The Plenum, and its constants	1, 2, 3, 4, 5, 20, 21
The solar system	6, 7, 8, 10, 11
The strong field	9, 24, 25, 26
Waves, and the bounds they set	12, 13, 15, 16, 17, 19, 22, 23
Galaxies and the expansion	14, 18, 27

The epistemic summary is generated

The closing SUMMARY OF EPISTEMIC STATUS is built from the same register the claims are, so it cannot drift from them, and every line of it carries its claim number. The four results the framework establishes that are not separately numbered claims are carried in it and labelled as such, so nothing the summary used to say was lost.

Reporting one instrument, one claim, or the register alone

python3 verify_neogravity.py	the verification
python3 verify_neogravity.py --register	the claim register, then stop
python3 verify_neogravity.py --family strong	report one instrument
python3 verify_neogravity.py --claim 9	report one claim
python3 verify_neogravity.py --json	register and results, machine-readable

These select what is **reported**. The program always **computes** all twenty-seven, because the later claims use the earlier ones’ values, and a filtered run prints a banner saying it is not the verification. --json is what the verification page on NeoGravity.org is generated from, so that page cannot disagree with the program.

Purpose

Every central mathematical claim in the series is reduced here to executable code. A reviewer can verify the derivation chain rather than checking algebra by hand.

The suite is deliberately adversarial toward its own theory. It marks [WRONG] on phrasings that overclaim, states which quantity is fitted rather than derived, and identifies the results that are *not* independent confirmation. It exits non-zero if any check fails.

Running

```
pip install sympy numpy
python3 verify_neogravity.py
```

Output is reproduced verbatim in `sample_output.txt`. One file is all that ships: the four per-paper scripts written in September 2026 have been folded in as Claims 6, 13 and 14 rather than deposited separately.

Conventions

Term	Definition
Ether Plenum (The Modern Ether)	The continuous, Lorentz-invariant fluid proposed to fill space: quantum-vacuum zero-point energy, the Higgs field vacuum expectation value, and the universal electromagnetic background.
Radiative Undertow	The inward -directed reaction load returned by the Ether Plenum on matter, under Newton’s Third Law, in response to the outward transport of mass-energy. Identified with gravitational attraction.
Ether Density (ρ_{ether})	Local mass-energy density of the Plenum. Increases toward a gravitating mass.
Ambient density	$\rho_0 = 4.645 \times 10^{-31} \text{ kg m}^{-3}$, from the CMB temperature by Stefan-Boltzmann and $E = mc^2$.
Equation of state	$w = 1/3$, derived from the traceless stress tensor of any massless field. Not fitted.
Equilibrium profile	$\rho(r) = \rho_0 \exp[3GM/(c^2r)]$, from hydrostatic balance with $w = 1/3$.
Refractive index	$n = (\rho/\rho_0)^k$ with $k = 2/3$, exactly. Equivalently $n(r) = \exp(2GM/rc^2)$. A denser medium slows light, as glass does. The linear form $n \approx 1 + 2GM/rc^2$ is its first-order expansion, not the framework’s index; no prediction may rest on the expansion.

The kinematic-dynamic boundary. A plenum account reproduces the *kinematics* of general relativity — how signals propagate through a given background — without

thereby reproducing its *dynamics*, what determines that background. Every result below is a statement about propagation. Every open problem is a statement about sourcing. Passing every kinematic test is necessary and not sufficient.

What is checked

#	Claim	Method
1	Hydrostatic equilibrium of the Plenum	SymPy dsolve on a first-order ODE
2	The exponential is not a $1/r$ power law	Series expansion at large r
3	$w = 1/3$, forced by the traceless stress tensor	Trace of the stress tensor of a massless field
4	Ambient density ρ_0 from the CMB	Stefan-Boltzmann and $E = mc^2$
5	Calibration of the optical exponent $k = 2/3$	First-order coefficient solved against the measured deflection
6	Second-order solar-limb deflection, $0.73 \mu\text{as}$	Taylor expansion of both indices; closed form $\pi \cdot \Delta a_2 \cdot x^2$
7	Mercury perihelion advance	Secular advance formula
8	Gravitational redshift and Shapiro delay	Pound-Rebka tower, $z = gh/c^2$; Shapiro $\Delta t = (4GM/c^3) \cdot \ln(4r_1 r_2/b^2)$
9	Photon-capture diameter $b_c = 2e \cdot GM/c^2$	Stationary point of $b(r) = n(r) \cdot r$
10	Frame dragging: topology, and R_{eff} as definitional	Explicit statement of what is matched
11	Two constraints on the Undertow	Luminosity-to-mass ratios against MICROSCOPE; radiation pressure against gravity
12	Hellings-Downs from the transverse-traceless family	Correlation evaluated across separation
13	The compression residue κ_L , bounded	Kramer et al. (2021) double-pulsar precision
14	The withdrawn vorticity account of galactic rotation	Recorded as withdrawn

#	Claim	Method
15	A potential body force sources the compression branch only	$\text{curl grad } \Phi = 0$, symbolically
16	The relaxational source rule $\kappa_L = \kappa_0/(\omega\tau_s)^2$, a MODEL	Settling time bounded by the double pulsar
17	The wave-sector energy is coupling-set field energy, not the Plenum's inertia	Ratio to the Plenum's inertial energy computed; recorded as a postulate
18	The relaxation window's top is the Hubble time; a galaxy lies below it	$1/H_0$ against the window; the galactic Deborah crossing
19	The transverse source at the double pulsar, and what P4 viscous coupling gives	Required solenoidal force against Stokes entrainment across the window
20	$\beta = 1$, the nonlinearity parameter	Read off the PPN perihelion coefficient with $\gamma = 1$
21	Shear modulus $\mu(\rho_0) = \rho_0 c^2$, at the ambient density only	$c_T = \sqrt{\mu/\rho_0} = c$, solved
22	Finding A6.1: steady amplitudes are viscosity-independent	The steady Stokes field of a rotating sphere carries no η
23	Finding A6.2: the barotropic law annihilates the baroclinic source	$\nabla\rho \times \nabla p \equiv 0$ for $p = p(\rho)$, symbolically
24	Finding A8.1: the inflow is strictly subluminal, so there is no acoustic horizon	$v/c = \sqrt{1 - e^{(-2x)}}$; 0.795 at the capture surface; the limit at $r \rightarrow 0$
25	The areal radius and the wormhole rival. Recorded, not settled	Stationary point of $R(r) = r \cdot e^{(m/r)}$
26	Finding A12.1: the constitutive arrow, so no time-reversed threshold	The Maxwell element under $t \rightarrow -t$, for any $\tau > 0$
27	Finding A13.1: the conformal mapping and the three inherited tests	Null rays independent of $a(\eta)$; the $(1+z)$ scalings
—	The claim count itself	

#	Claim	Method
		The program counts its own claims and refuses to finish if the number has drifted
—	Dimensional consistency	The density-gradient law reduced to Newton

Epistemic status

DERIVED — follows from the postulates with no free parameters:

- Ambient density $\rho_0 = 4.645 \times 10^{-31} \text{ kg m}^{-3}$, from the CMB
- Inverse-square form of the Undertow (spherical shell geometry in $\mathbb{R}^3$)
- **The equation of state $w = 1/3$** , forced by the traceless stress tensor of any massless field, not fitted
- Equilibrium density profile and refractive index form
- The temporal response coefficient, from energy conservation alone
- The density-gradient law $a = -(c^2/3)\nabla \ln \rho$, which contains no mass, no separation and no gravitational constant, and recovers Newton exactly
- Mercury's perihelion advance, $42.98''/\text{century}$ against 42.98 ± 0.04
- Gravitational redshift
- Shapiro time delay
- The transverse-traceless wave sector, and with it the Hellings-Downs inter-pulsar correlation
- Rotational frame dragging

FITTED — one quantity:

- $k = 2/3$, the optical exponent, equivalently the parametrized post-Newtonian parameter γ , which every metric theory of gravitation carries and none derives from first principles. Calibrated on the measured solar light deflection.

A second constant governing the frame-dragging amplitude was fitted in Paper 12 and has since been retired as definitional: R_{eff} is defined by $\Omega R_{\text{eff}}^3 = 2GJ/c^2$ and carries no independent content, so the ratio it is checked against returns 1 by construction.

PREDICTED, AND NOT YET MEASURABLE:

- A photon-capture diameter $b_c = 2e \cdot GM/c^2 \approx 5.437 \text{ GM}/c^2$ against the Schwarzschild $3\sqrt{3} \approx 5.196$ — a **4.63 %** excess, mass-independent, at zeroth order in spin
- A second-order solar-limb deflection differing from general relativity by **0.73 μas** ($0.7298 = \pi \Delta a_2 \cdot x^2$), roughly thirty times below current astrometric precision

BOUNDED, NOT DERIVED:

- The compression residue $\kappa_L < 1.3 \times 10^{-4}$, from the double pulsar's own published precision. Order-counting puts it near 4×10^{-6} , some thirty times inside that bound. **What the series has not derived is the coefficient itself.** No computation of this coefficient from the continuum equations exists; producing one is the outstanding task, and the elastic reading is withdrawn if the computed value exceeds the bound
- The scalar-longitudinal amplitude of the nanohertz background, below 4.2×10^{-17} (Wu et al. 2022) — a second and independent handle, four decades lower in frequency

TOPOLOGY ONLY — shared functional form, physically distinct cause:

- The frame-dragging dipole $\sin(\theta)/r^2$

CONSTRAINT — required by observation, not supplied by the theory:

- The coupling tracks total mass-energy, not thermal luminosity. Comparing the Sun's luminosity-to-mass ratio with the Earth's total infrared emission — the conservative of the two defensible readings, and the one the series' FAQ quotes — leaves a disparity near 9,400, against MICROSCOPE's 10^{-15} bound on composition-dependent free fall. Taking the Earth's own internal heat instead gives 2.4×10^7 ; the suite prints both and argues from the smaller

What changed from version 4.1

Taken from the program's own header, so that the two cannot drift apart.

The compression residue is no longer bounded-and-not-derived. Claim 13 now derives it from P1 to P5 with nothing added – the orbit's loss to the compression branch is the work of that branch's own reaction field on the bodies, which the reading law fixes – and the answer, $(c/c_L)^{7/18} = 9.3e-3$ of Peters-Mathews for the Newtonian body force and 0.23 for the drive that holds the full profile, lies ABOVE the double pulsar's bound. That meets the retroactive-falsification clause the series set itself, so the formulation as first written fails on that branch, and the successor formulation narrows the reading law: matter reads the settled profile and not the compression wave. The order-counting estimate near four parts per million is withdrawn and printed as withdrawn.

Claim 16, the relaxational source rule, is WITHDRAWN for the same run of work: the settling time it needs is supplied neither by the postulates (the Plenum settles a compression across the orbit in 2.27 s, four orders short) nor by the sourcing without a lunar lag that laser ranging excludes by ten orders.

Open problems 4, 5 and 6 and the suite-task list are brought level with both.

What changed from version 4.0

A reading of the corpus on 14 September found the claim list short in a way that mattered. Every numbered **Finding** in the strong-field and cosmological papers was outside it, and two rows of the Status Register (*The Central Roadmap*, Paper 20, Table 3) marked **Derived** had no check behind them. Eight claims were added, 20 to 27. Each recomputes something the papers already deposit; none is new physics.

The one that matters most is **Claim 24, Finding A8.1**. Paper 28 calls it “the sharpest internal discipline here” and marks it *Derived and verified*, and the suite did not verify it. The framework’s inflow speed $\sqrt{1 - e^{(-2x)}}$ passes $0.795c$ at the capture surface and approaches c only as $r \rightarrow 0$: strictly subluminal everywhere, so there is no sonic surface and no acoustic horizon. The standard river reaches c at r_s ; this profile never does.

Claim 25 records a rival rather than settling it. The areal radius of the exponential metric has a stationary point at $r = GM/c^2$, inside the capture surface. That is the Boonserm–Ngampitipan–Simpson–Visser wormhole reading. Whether it is a traversable throat is a question about stress-energy, which this program cannot adjudicate; it computes the mathematics and says so plainly. Paper 28 records the rival as standing, and Paper 29 is the argument against it.

Two corrections to claims version 4.0 already had, both found by re-reading the gate against the corpus rather than against itself:

- **Claim 8** was titled “Gravitational redshift and Shapiro delay” and computed no Shapiro delay at all. It also checked the *solar surface* redshift against a “measured 2.12×10^{-6} ” that no paper in the corpus states. Both repaired: the redshift is now computed over Pound and Rebka’s own tower and reproduces the corpus’s 2.459×10^{-15} , the Shapiro delay reproduces the corpus’s $232.6 \mu\text{s}$, and the solar figure is kept as context with its false label withdrawn.
- **Claim 18** used 220 km/s for a galaxy’s speed, which is the *Sun’s* figure from Paper 39’s Table 1, and printed $4.21 \times 10^{15} \text{ s}$ beside a citation of that paper’s 4.6×10^{15} . Corrected to the galaxy’s own 200 km/s , which reproduces $4.6 \times 10^{15} \text{ s}$.

The count is now asserted by the program. The forty papers quote the number of claims in their Data and Code Availability sections, so the number is a fact about this program that the corpus repeats. `verify_neogravity.py` counts its own claims, checks that they run 1 to N without a gap, and fails if either has drifted.

Nothing version 4.0 checked was dropped; the claim list is a superset.

Corrections enforced here

1. **$\exp(A/r)$ is not a $1/r$ power law.** The $1/r$ dependence sits in the exponent; a $1/r$ perturbation appears only at first order. Manuscripts claiming the integration “produces a $1/r$ profile” are wrong as phrased.
2. **The exponential index is exact, and the linear form is only its expansion.** A prediction resting on $n \approx 1 + 2GM/rc^2$ is void: a linear index has no stationary point in $b(r) = n(r) \cdot r$ and therefore yields no photon capture and no shadow at all.
3. **Frame dragging is not “algebraically identical” to Stokes flow.** Stokes creeping flow requires viscosity and a no-slip boundary; Kerr is a vacuum solution. They share dipole topology only.
4. **Earth’s moment of inertia is $I = 0.3307 M R^2$, not $(2/5)M R^2$.** Earth is centrally condensed; the uniform-sphere value overstates J by 21 % and yields an incorrect R_{eff} near 5.24 km. Any effective radius must be quoted with the convention that fixes it, or it is not a number.
5. **Light deflection is not independent confirmation.** The optical exponent was calibrated on that measurement. The Shapiro delay is a same-parameter consistency check rather than an independent success, and the redshift is a temporal input to the calibration. The genuinely independent success is Mercury’s perihelion advance.
6. **Radiation pressure is not the Undertow.** Measured solar radiation pressure is 6.1×10^{13} times too weak, and points outward rather than inward.

Constants

CODATA 2022 and IAU 2015 nominal solar values, declared at the top of the script. Every constant used is either exactly defined or unchanged from CODATA 2018.

Citation

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